Co-Packaged Optics

Architecture, adoption and value capture in AI-scale networks

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Architecture, adoption and value capture in AI-scale networks

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> **Research disclosure**
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> This report is for informational and educational purposes only. It is not investment advice, a solicitation, or a recommendation to buy or sell any security. Conclusions are evidence-gated and may change as customer, yield, qualification and supplier-economics data become public.

1.1 Executive conclusion

The central view: switch-side CPO at 200G/lane has the strongest currently disclosed commercial signal for Ethernet scale-out. NVIDIA states that Spectrum-X Ethernet Photonics, its CPO-based 200G-SerDes switch system, is in production; Broadcom has announced a third-generation 200G/lane CPO route and a 102.4T CPO demonstration path. These are important product-route signals, not independently reconciled accepted-unit, yield or margin disclosures (NVIDIA 2026c; Broadcom 2025b, 2026).

Decision	Current view	Evidence grade	What would change it
First commercial domain	Ethernet scale-out switching	Inference	A named, repeat accelerator-side optical-I/O deployment with accepted volumes
Commercial proof	2026–2027 is the key verification window	Forecast	Customer SKU, accepted units and repeat shipment evidence below expectations

Decision	Current view	Evidence grade	What would change it
Architecture pressure	200G/lane increasingly favours a short electrical boundary; LPO/NPO remain real countercases	Inference	Matched LPO/NPO field results at similar lane rate, power and service cost
Technical/control leaders	NVIDIA: integrated platform; Broadcom: merchant CPO route; TSMC: packaging control; Coherent/Lumentum: component-stack exposure	Inference	Disclosed supplier share, yield, ASP and customer adoption data
Investable conclusion	Named views are provisional; no CPO-specific profit-pool leader is yet proven	Open question	Attributable revenue, margin, qualified yield and warranty evidence

1.2 Named investment views

- **NVIDIA — positive strategic exposure, medium confidence.** The company's integrated platform and declared production status create the clearest switch-CPO commercialization route. The unresolved issue is how much incremental CPO economics accrue to NVIDIA versus the supply chain (NVIDIA 2026c).
- **Broadcom — positive enabling exposure, medium confidence.** Broadcom has the clearest merchant switch-silicon/CPO product-definition route, but production customer, supplier-content and field-reliability evidence remain limited in public disclosure (Broadcom 2025b, 2026).
- **Coherent — constructive component-stack exposure, medium confidence.** Coherent's demonstrations span silicon photonics, external lasers, VCSELs, InP modulators and packaging. Demonstrated breadth is not proof of design wins or CPO margins (Coherent 2026c).
- **Lumentum — watch for external-laser conversion, medium confidence.** External-laser content can be a valuable CPO input, but public evidence must still establish quantity, customer concentration and margin capture.
- **Marvell and TSMC — strategic watchlist, not a CPO-specific call.** Marvell's accelerator optical-I/O route and TSMC's advanced-packaging position matter, but their CPO profit attribution is not yet publicly separable.

1.3 The investable question

The report does not ask whether photonics is technically useful. It asks a stricter sequence of questions:

1. Is the electrical channel, power density or front-panel density constraint binding enough that an integrated optical route is required?
2. Can that route be manufactured, qualified and serviced at a cost that survives the yield penalty of deeper integration?
3. Which layer holds the scarce control point after the platform owner, component supplier, OSAT, laser supplier and operator divide the economics?
4. Is any public company's incremental profit opportunity large enough to matter relative to its existing revenue, gross margin, capital intensity and valuation?

The answer to the first question is becoming more favourable at 200G/lane in Ethernet scale-out. The answers to the remaining three are not yet public facts. That distinction is central: a persuasive technology roadmap can coexist with an uninvestable supplier profit pool.

> **i** Current stance in one page
>
> **Architecture:** CPO is most credible where 200G/lane makes the host electrical boundary and optical-interface power binding. It is not the default answer for every rack-scale link.
> **Timing:** 2026–2027 is the commercial-proof window. The base case requires customer-confirmed, repeat paid production; a vendor production statement alone is necessary but insufficient.
> **Value capture:** The potential beneficiaries are platform integrators, qualified-engine/PIC designers, external-laser suppliers and advanced package/test routes. No public record yet identifies their realised share, qualified yield or margin.
> **Portfolio interpretation:** NVIDIA and Broadcom have the clearest system/product routes; Coherent and Lumentum have the clearest optical-content routes; Marvell and TSMC are strategic optionality/control-point cases. All six remain evidence-gated rather than valuation calls.

1.4 What would make the view materially stronger

The highest-value public evidence would be a record that joins five currently separate disclosures: a named CPO SKU; accepted unit or port range; repeat shipment; a supplier or content boundary; and a qualified-yield or service metric. The next-best record is a matched CPO/LPO/NPO comparison that includes the same throughput, reach, optical boundary, facility-power method and replacement policy. Until then, the report treats product launches, lab demonstrations and ecosystem partnerships as signals, not as complete proof.

The most relevant countercase is not "no optics." It is continuing innovation in retimed pluggables, linear-drive optics, linear pluggable optics, active electrical cables and near-packaged routes. An architecture becomes durable only if it improves delivered cost, service and power together. The OIF's energy-efficiency work and the IEEE 802.3dj lane-rate process show why the industry is working the boundary from both electrical and optical directions (Optical Internetworking Forum 2024; IEEE 802.3 2024).

1.5 How to use this report

Start with the company cases for the investment call, then test it against the architecture, manufacturing and value-pool chapters. The required sequence is: define the product boundary; compare CPO with the strongest LPO/NPO/pluggable countercase at the same lane rate; identify the qualified control point; then test whether its profit can matter to the company. Every conclusion names the observation that can change it.

1.6 What changed this quarter

1. NVIDIA publicly described Spectrum-X Ethernet Photonics as a CPO-based 200G-SerDes system "now in production" (NVIDIA 2026c).
2. Broadcom presented a 102.4T CPO route alongside 200G/lane connectivity at OFC 2026 (Broadcom 2026).
3. Coherent demonstrated multiple CPO technology paths, including 6.4T socketed CPO and 400G-per-lane InP-on-silicon work (Coherent 2026c).

See the change log for the evidence boundary and version history.

1.7 Read the report

For the concise, decision-memo version, read the CPO investment brief.

Start with the decision framework, or go directly to architecture, adoption scenarios, or the company investment cases.

Download the print edition (PDF)

2 Decision framework

2.1 Research question

> At what bandwidth, topology and system scale does CPO become economically preferable or technically necessary relative to retimed pluggables, linear pluggables, NPO, retimers and active electrical cables—and which suppliers gain or lose profit content as the transition occurs?

The report centres on **scale-out Ethernet optical engines/PICs**. Switch-side CPO and accelerator-side optical I/O are related but different product, customer and value-pool questions. They are not aggregated into a single adoption forecast.

2.2 Claim convention

Label	Meaning	Publication rule
Fact	Direct statement or measurement from an attributable public source	Cite the source and keep the boundary intact
Inference	Analytical conclusion from multiple facts	State the reasoning and key limitation
Forecast	Timing, adoption or economics expectation	State horizon, scenario and falsification trigger
Open question	Decision-critical data not publicly available	Do not fill the gap with an unsupported estimate

2.3 Evidence standard

The evidentiary order is: standards and filings; peer-reviewed/conference research; official product specifications; customer/supplier evidence; management commentary; and market commentary. A vendor claim can establish its own roadmap; it cannot independently prove market-wide yield, comparative power, field reliability or supplier profit capture.

> **Public-data boundary**
>
> The public book contains only claims that can be traced to an original public source. Restricted papers, raw web archives, private notes, primary research, unpublished models and copyrighted

PDFs stay outside the site.

2.4 Decision gates

1. **Technical gate:** electrical reach, optical budget, thermal and package feasibility at the required lane rate.
2. **Manufacturing gate:** qualified yield after fibre attach, test, rework, reflow and stress exposure.
3. **System gate:** serviceability, failure isolation, host integration and customer acceptance.
4. **Economic gate:** attributable content, realised ASP, gross margin, warranty and capital intensity.

The thesis is not complete until all four gates are evidenced. A compelling demo can clear the first gate without clearing the other three.

2.5 The unit of analysis

The report compares architectures at a stated boundary. A claim about an optical component is not automatically a claim about a switch, and a claim about a platform is not automatically a claim about a supplier's revenue. The minimum comparison unit is a delivered port at a stated lane rate, reach and service model. It includes the optics and the incremental electrical, thermal, conversion and replacement burden required to make the port useful to an operator.

This prevents three common analytical mistakes. First, a low pJ/bit component result can omit the laser, receiver, host SerDes, power conversion or cooling. Second, a production announcement can cover a family of switches, including pluggable configurations, rather than the CPO SKU in question. Third, a supplier partnership can establish capacity access without establishing product allocation, content share or price.

Boundary control	Required disclosure	Why it matters
Architecture	CPO, NPO, LPO, retimed module or accelerator optical I/O	Prevents comparing different products as if they were substitutes
System	Switch or accelerator, port count, lane rate and reach	Defines the electrical and thermal problem
Commercial	Named SKU, customer, shipment state and repetition	Separates product language from adoption
Economic	Supplier content, accepted yield, ASP, margin and warranty	Separates technical relevance from profit capture

The hierarchy and claim labels above apply to every chapter: lower-tier sources can identify a diligence question, but cannot settle a higher-tier economic claim. A supplier demonstration can establish a device route; only customer, qualified-manufacturer or contractual evidence can establish accepted volume and allocation.

3 What CPO solves

3.1 The physical problem

At higher lane rates, a long electrical path from switch ASIC through package escape, PCB, connector and faceplate module increasingly consumes channel margin and power. CPO moves the optical engine close to the host ASIC; the potential gain is a much shorter high-speed electrical path. The cost is a tighter integration, thermal, yield and serviceability problem.

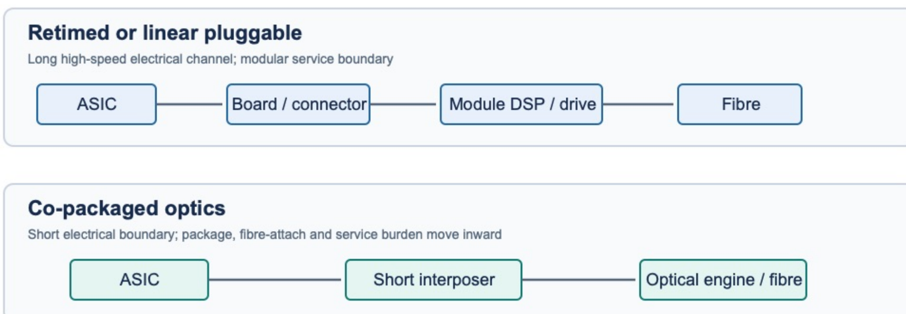

Figure 3.1: Original analytical diagram comparing the electrical and service boundaries of retimed or linear pluggable optics with co-packaged optics. The diagram synthesises the architecture boundary described by the OIF's co-packaged-module framework and energy-efficiency requirements; it is not a measured power or cost comparison (Optical Internetworking Forum 2023, 2024).

3.2 The trade-off is not simply power

Constraint	Why CPO can help	New burden CPO creates
Electrical reach	Removes much of the faceplate electrical channel	Tighter package and SerDes co-design
Power and density	Can reduce retiming/DSP burden and free faceplate space	Thermal density moves closer to the ASIC
Optical integration	Enables short, dense engine placement	Fibre attach and optical test become system-level operations
Service	Potential for known-good subassemblies/ELS architectures	A failed engine is harder to replace than a pluggable

Constraint	Why CPO can help	New burden CPO creates
Economics	May shift content toward PICs, lasers, packaging and test	Final yield, rework and warranty can erase the gain

3.3 Where the electrical boundary matters

The relevant constraint is not cable length alone. The channel includes host-package escape, board routing, vias, connectors, the module interface and the equalisation burden required to keep the transmitter and receiver inside their error and eye-quality budgets. At a low enough loss and a mature enough module ecosystem, retimed pluggables retain the advantage of modular service. At a difficult enough channel, the price of preserving that modularity rises through DSP, retiming, power, front-panel thermal load and design complexity.

The OIF's energy-efficiency and co-packaged-module work frame this as a system architecture choice, not a vendor-specific claim (Optical Internetworking Forum 2024, 2023).

Signal path	Electrical boundary	Service boundary	Core trade-off
Retimed or linear pluggable	ASIC → package/board → connector → module	Module at faceplate	Preserves modular replacement but consumes channel margin and faceplate power.
CPO	ASIC → short interposer → optical engine	Engine/package assembly	Shortens the electrical path but concentrates package, fibre-attach and service risk.

3.4 Power is necessary, but not sufficient

A 102.4T thought experiment illustrates the distinction. With 64 x 1.6T ports, an assumed 7.2 W CPO optical interface per port produces 460.8 W of direct interface power; a 24 W fully retimed interface produces 1,536 W. These are scenario calculations, not a measured chassis comparison. They omit the switch core, endpoint optics and common chassis loads and therefore should not be read as a complete facility-power claim. Under the same assumptions, the CPO case is approximately 70% lower than the fully retimed interface case, while the gap versus an 8 W LPO case is only 10% (NVIDIA 2025b; Marvell 2025).

That last comparison is the investment challenge. If a linear or near-packaged alternative can preserve serviceability while remaining near CPO on power, CPO must earn its integration penalty through density, electrical reach, thermal design or a better qualified-cost curve.

Question	CPO-friendly answer	Countercase that would weaken the thesis
Electrical margin	Host channel is the limiting link budget	LPO/advanced pluggables demonstrate comparable margin at the same lane rate
Optical power	Retiming/DSP burden is material	Linear route remains close in full-system power
Density	Faceplate and routing constrain system scale	Modular optics fit within thermal and mechanical envelope
Operations	External laser/connector route isolates failures	Replacement, MTTR and spare cost favour modular optics

4 Architecture contest

4.1 Five architectures, five different constraints

Architecture	Electrical path	DSP / retimer	Field replacement	Current read
Retimed pluggable	Long	In module	High	Mature baseline; power and density become difficult
Linear pluggable optics (LPO)	Long	Reduced or omitted	High	Strong countercase where host-channel margin holds
Near-packaged optics (NPO)	Shorter	Architecture dependent	Moderate	Intermediate route with fewer CPO service penalties
Switch-side CPO	Very short	Often relocated/reduced	Low	Strong 200G/lane commercial signal; production economics unproven
Accelerator optical I/O	Package/die-scale	Architecture dependent	Very low	Separate longer-duration scale-up opportunity

4.2 Decision matrix

Test	Pluggable/LPO	NPO	CPO
Best when	Flexibility and field replacement dominate	Electrical channel is tight but serviceability matters	Density/electrical reach is the binding constraint
Must prove	Host margin, interoperability, error/power distribution	Package integration and replacement method	Final yield, fibre attach, qualification and service economics

Test	Pluggable/LPO	NPO	CPO
Main disproof	Cannot meet channel/power requirements	Does not improve TCO versus pluggable	Yield/service penalty outweighs electrical/power gain

4.3 Architecture is a portfolio of trade-offs

Retimed pluggables move electrical complexity into the module and retain the cleanest replacement model. LPO removes or reduces module DSP complexity but demands a disciplined host-channel and interoperability environment. NPO is a useful category when the electrical path must shrink but the operator still wants an intermediate replacement boundary. CPO moves the optical engine closest to the ASIC; it has the greatest ability to compress the high-speed electrical path and the greatest exposure to package, attach, qualification and field-service economics.

Accelerator optical I/O should not be used to inflate switch-side CPO adoption. Its product boundary is a scale-up fabric or chiplet/interposer architecture, with different buyers, host integration, failure domain and supplier set. Marvell's silicon-photonics light-engine announcement and Intel's optical-I/O-chiplet work demonstrate why this domain matters strategically, but they do not establish a switch-CPO volume curve (Marvell 2025; Intel 2024).

Decision gate	If yes	If no
Does the host electrical channel hold at the required rate and reach?	Test whether module power/density remains acceptable.	Evaluate LPO/NPO or a shorter electrical boundary.
Is module power/density acceptable?	Retimed or linear pluggable remains viable.	Evaluate LPO/NPO.
Can a serviceable intermediate boundary work?	Consider NPO or a detachable-engine route.	Evaluate switch-side CPO.
Are yield and service economics qualified?	Commercial CPO candidate.	Architecture remains unproven.

4.4 The 200G and 400G distinction

At 200G/lane, product-route evidence is now specific enough to justify serious diligence: Broadcom describes the BCM78919 as a 102.4T multilayer CPO switch with 200G SerDes, and NVIDIA describes a 200G-SerDes Spectrum-X Photonics route (Broadcom 2025e; NVIDIA 2026c). Neither disclosure supplies a matched pluggable/LPO TCO, a final yield waterfall or accepted unit data.

At 400G/lane, technology demonstrators show why the boundary is being investigated but not why a CPO winner has been established. The Zhou paper models a 212.5-GBaud linear-drive environment; the disclosed result depends strongly on the assumed bump-to-bump loss (Zhou et al. 2026). The correct takeaway is conditional: as the electrical loss budget tightens, deep integration becomes more valuable. The incorrect takeaway is that a universal loss threshold automatically establishes CPO economics.

Investment implication

Lane rate is a trigger for diligence, not a valuation input. A supplier needs both an architecture advantage and evidence that it controls a profitable, qualified portion of the implementation.

4.5 Comparison protocol for technical claims

The report applies a fixed protocol before comparing a CPO claim with a pluggable, LPO or NPO claim. The lane rate and aggregate bandwidth must match; a 100G/lane result cannot settle a 200G/lane question. The reach, fibre type, FEC threshold, modulation format and electrical channel loss must be stated. The power boundary must state whether it includes laser, driver, receiver, host SerDes, module DSP, voltage conversion and cooling. Finally, the operational boundary must identify the replaceable element and the recovery procedure. If these fields are absent, the source can illustrate a mechanism but not establish an architecture winner.

This is especially important because marketing labels conceal different engineering allocations. A "linear" solution can still depend on particular host equalisation, connector quality, cable/reach limits and receiver sensitivity. A "co-packaged" solution can use different engine counts, laser distribution, package design and replacement architectures. The reported pJ/bit number may be correct within its boundary yet unhelpful for the investment question if the omitted burden is material.

Comparison field	Minimum disclosure	Why an omission changes the conclusion
Lane and modulation	Gb/s per lane, baud rate, PAM/coherent mode	Determines electrical and optical bandwidth burden
Link boundary	Die-to-die, host-to-module, fibre reach and connector count	Prevents a short lab path from representing a deployed link
Error criterion	BER/TDECQ/FEC threshold and equalisation method	Defines whether performance is actually comparable
Power boundary	Included chips, laser, conversion and cooling	Prevents low component power from becoming low system power
Service boundary	Replaceable module, laser, fibre array or engine	Determines operating cost and downtime risk
Manufacturing boundary	Die, engine, board or final system acceptance	Determines whether yield claims are decision-useful

The report treats a comparison as **matched** only when enough of these fields are public. That is a high bar, and the present record does not clear it for a full 102.4T CPO/LPO/NPO/retimed TCO comparison. The absence itself is informative: it signals that investment conclusions should be architecture- and timing-oriented rather than a precise cost or earnings claim.

4.6 NPO deserves separate diligence

NPO is often placed between pluggables and CPO on a simple integration spectrum. That is useful but incomplete. A near-packaged approach can alter the electrical path and thermal layout without inheriting every CPO service boundary; it may also create a new package, connector or cable-management problem. IEEE 400G-per-lane study-group material explicitly treats NPO and CPO as distinct considerations at the next lane-rate boundary (China Mobile and IEEE 802.3 400G per Lane Study Group 2026).

The investor implication is that NPO is not merely a delayed CPO outcome. It can be a genuine substitute if it captures enough electrical/power benefit while preserving a better acceptance, replacement or supply-chain model. Any claim that CPO necessarily wins at 400G/lane must therefore be tested against NPO as well as LPO and advanced pluggables.

5 Scale-out optical engines and PIC design

5.1 The engine is a coupled system

An optical engine is not a single photonic chip. It is the coupled design of PIC, EIC/driver/TIA, laser architecture, fibre interface, thermal path, electrical package and test strategy. The most valuable technology is the one that improves the cost of a **qualified good engine**, not merely a laboratory metric.

5.2 Where durable economics could accrue

Control point	Potential advantage	Evidence still required
PIC/EIC co-design	Electrical-optical margin, area and power	Comparable yield, test time and customer qualification
External laser source	Serviceability and optical-power control	Content per switch, laser qualification and replacement economics
Fibre coupling/connector	Known-good-engine test and rework option	Mating life, loss distribution and automated throughput
Advanced packaging	Short channel and dense integration	Final yield, capital intensity, warranty and supplier margins
Test	Escape-rate control and lower field cost	Cycle time, coverage and cost per qualified engine

5.3 Scale-out remains the near-term focus

Coherent's OFC 2026 work illustrates the breadth of competing component routes: a socketed silicon-photonics CPO with external laser source, VCSEL-based CPO, and an InP-on-silicon 400G-per-lane modulator demonstration (Coherent 2026c). This supports technology optionality. It does not establish which route has the best qualified cost, customer acceptance or profit pool.

Working investment rule

Do not award a component supplier a structural CPO margin premium based on a device demo. Require evidence of design win, yield, attach/test burden, share of engine BOM, warranty allocation and repeat customer volume.

5.4 The design problem is an optimisation, not a parts list

An engine can use a silicon-photonics, InP, thin-film lithium-niobate or heterogeneous route; its commercial merit depends on how the choice interacts with electrical drive, laser delivery, coupling, package thermal paths and test. A PIC with an impressive modulator bandwidth can lose at the engine level if it requires expensive alignment, narrow thermal control, low probe coverage or a high field-failure burden. Conversely, an apparently less elegant architecture can win if it improves known-good-die test, replacement or manufacturing throughput.

Engine layer	Technical objective	Economic consequence	Public decision test
EIC / SerDes interface	Preserve signal margin with low drive energy	Determines host/package complexity	Matched electrical path and energy boundary
PIC	Modulate, multiplex, detect and route light	Determines die area, test and optical loss	Measured link performance at stated conditions
Laser / ELS	Deliver stable optical power	Can create serviceable external content	Laser count, delivered power and replacement allocation
Fibre interface	Couple channels reproducibly	Compounds final-engine yield	Alignment tolerance, loss distribution and mating evidence
Package and test	Protect, cool and screen the engine	Drives cost per accepted engine	First-pass yield, rework, coverage and qualification

The OIF ELSFP work is important because external-light architectures can separate the replaceable laser boundary from a deeply integrated optical engine (Optical Internetworking Forum 2025). This is not a free serviceability solution: laser distribution, connectors, optical power budgeting, contamination, qualification and operational spares become part of the system requirement.

5.5 Scale-out PIC thesis

The investable scale-out opportunity is a reproducible engine platform that carries its test coverage, laser architecture and attachment process across switch configurations. Coherent demonstrates route breadth; TSMC demonstrates that process integration can be a distinct control point (Coherent 2026c; TSMC 2024, 2026b). Public evidence still does not map qualified supplier share or margin.

6 Manufacturing, reliability and serviceability

6.1 The adoption bottleneck may be manufacturing

The technical case for short electrical paths is necessary but insufficient. In CPO, optical alignment, fibre attach, final package integration and qualification sit closer to the system critical path. IBM's full-module test vehicle and a recent heterogeneous-integration review both identify packaging, reliability and maintainability as coupled constraints, not afterthoughts (Knickerbocker et al. 2024; Gao et al. 2025).

6.2 Qualified-good-engine equation

$$\text{Qualified cost per engine} = \frac{\text{materials} + \text{assembly} + \text{test} + \text{rework} + \text{capital} + \text{warranty}}{\text{accepted engines}}$$

The numerator and denominator must be measured at the same boundary. A paper's coupling loss, a company's test-tool announcement or a one-time reflow result cannot by itself determine this value.

Manufacturing question	Why it matters	Public status
Fibre-attach first-pass yield	Can compound across many interfaces	Not broadly disclosed
Rework and retest yield	Determines whether known-good-engine concepts work economically	Not broadly disclosed
Environmental qualification	Converts a demo into a deployment candidate	Partial prototype evidence
Field replacement	Determines operational acceptance and warranty exposure	No comparable public evidence
Test time and coverage	Determines throughput and escape-rate economics	Not broadly disclosed

6.3 Serviceability is an architecture variable

External laser sources, detachable optical interfaces and socketed engines may reduce the operational penalty of deep integration. The OIF's ELSFP work and a detachable-chiplet-connector prototype show the relevant service boundary, but neither establishes a production field-service model (Optical Internetworking Forum 2025; Psaila et al. 2023).

6.4 Yield compounds across interfaces

The central manufacturing point is mathematical before it is proprietary. imec's development-run data show why optical-interface yield and alignment tolerance are material engineering variables; they are not final-engine production yield (imec 2024). If an engine requires many independent connections, the yield of the finished board is approximately the product of the connection yields, before adding die yield, attach, rework, burn-in and system acceptance. At 1,024 interfaces, a 99.90% assumed connection yield implies only about 35.9% all-good yield if no rework is allowed; achieving 95% board yield requires approximately 99.995% per-connection first-pass yield. These are illustrative calculations, not a disclosure about any supplier.

Assumed per-connection first-pass yield	288 interfaces	512 interfaces	1,024 interfaces
99.500%	23.6%	7.7%	0.6%
99.800%	56.2%	35.9%	12.9%
99.900%	75.0%	59.9%	35.9%
99.950%	86.6%	77.4%	59.9%

6.5 Test and service define the real failure domain

The difference between a replaceable laser module and a replaceable optical engine is commercially important. A field-replaceable ELS can isolate a laser failure, but it does not automatically make the PIC, fibre attach or package replaceable. A socket, detachable fibre interface or known-good subassembly can improve serviceability, but it introduces its own insertion-loss, contamination, connector-life and handling requirements. The relevant operator metric is not only power; it is recovery time, spare inventory, field procedure, warranty ownership and blast radius.

Teradyne's discussion of silicon-photonics test challenges illustrates the industry direction: wafer, die, package and system test must be linked rather than treated as a conventional module-only workflow (Teradyne 2025). Public data do not yet disclose the actual test time, escape rate or cost per qualified engine for a CPO platform.

6.6 What would clear the gate

A credible manufacturing upgrade would disclose a defined product boundary and at least three of the following: fibre-attach yield distribution; rework route and recovery rate; environmental qualification sample/condition; optical/electrical test coverage; accepted-engine yield; field-replacement procedure; warranty allocation; or multi-customer production repetition. A lab coupler result is useful engineering evidence, but it is not a manufacturing business model.

6.7 Reliability is a system property

Thermal, optical and mechanical stresses interact in a CPO assembly. IBM's work reports full-module reflow and stress testing, while its test-vehicle boundary still falls short of production-lot yield and fleet reliability (Knickerbocker et al. 2024). A separated laser does not remove the PIC, interposer, fibre-interface or connector reliability burden. The report distinguishes **device** reliability, **assembly** reliability and **operational** reliability; only the last establishes whether an operator can isolate and repair the failure domain at acceptable cost.

Reliability layer	Typical evidence	What a positive result means	What remains unknown
Device	Laser/PIC ageing, thermal sweep, optical output	The tested component has a defined stress response	Final-package coupling, system use and field distribution
Assembly	Connector mating, reflow, thermal cycling, humidity	The test vehicle survives a stated integration exposure	Production variation, rework and acceptance yield
Product	Qualification, burn-in, acceptance test	A product boundary met a specified standard	Customer volume and in-field return distribution
Operation	MTTR, spares, swap procedure, warranty	The operator can manage the failure domain	Economics across the installed base

6.8 The cost of a late failure

The reason yield and service are central is the value of the assembly at the point of failure. A bare photonic die failure can be screened relatively early. A fibre-attach defect after the PIC, EIC, advanced package, substrate and laser distribution have been added consumes much more value. A field failure can add logistics, downtime, diagnosis, replacement inventory and warranty exposure. This creates a non-linear economic outcome: small changes in attach yield, test coverage or reworkability can be more valuable than a modest improvement in component power.

That logic is not a claim that CPO is uneconomic. It is a guide to what the winning implementation must demonstrate. A supplier that can improve test-before-integration, passive/automated attach, known-good-engine screening, detachable service boundaries or final qualification could capture more durable value than a supplier whose device is marginally faster but operationally hard to deploy.

7 Adoption timing and scenarios

7.1 Separate commercial proof from meaningful adoption

Commercial proof means repeat paid production from two independent customers, or sustained/repeated production by one major customer, with a clear CPO product boundary. **Meaningful adoption** means a disclosed unit/port denominator large enough to assess the technology's economic relevance. The latter is not currently quantifiable from the public record.

Scenario	2026–2027	2028–2030	Key evidence required
Bear	CPO stays concentrated in selected systems; pluggables, LPO, AECs and NPO improve	Service/yield penalty prevents broad scale-out adoption	Customer deployments fail to expand; low qualified yield or high service burden
Base	Switch-side 200G/lane CPO gains repeat production proof	Adoption remains concentrated in highest-bandwidth fabrics	Accepted-unit range, repeated shipments, final-yield waterfall and matched TCO
Bull	Qualified CPO proves repeatable across leading platforms	CPO expands into wider scale-out and selected scale-up domains	Strong field reliability, service solution and disclosed supplier economics

Forecast: 2026–2027 is the central commercial-proof window. NVIDIA's current production statement is the strongest positive public signal; it does not settle accepted units, customer mix, comparative economics or supplier profit capture (NVIDIA 2026c).

7.2 Falsification checklist

Downgrade the base scenario if any of the following occur:

1. Named customers deploy 200G/lane LPO or advanced pluggables with similar power and service cost.
2. CPO production announcements resolve to evaluation quantities or non-CPO SKUs.
3. Final-engine yield/rework remains poor after qualification.
4. Contracts show CPO component content is low-margin or exposed to rapid price-down.

7.3 A gated adoption timeline

This is a commercial-proof framework, not a market-share forecast. It can be upgraded or downgraded as customer and shipment evidence appears.

Architecture / domain	2026 public read	2027 base condition	2028–2030 condition	What would invalidate the path
Retimed and advanced pluggables	Mature baseline	Continues where flexibility dominates	Remains strong at higher lane rates if power/thermal holds	Material inability to meet electrical/power requirement
200G LPO	Important countercase	Needs qualified, interoperable system evidence	Could take share if service and power remain close to CPO	Host-channel margin or field performance breaks
200G switch CPO	Strongest disclosed product/production signal	Named SKU, accepted units and repeat shipment	Broader scale-out only if yield and service qualify	Production resolves to evaluation quantity or uneconomic service
400G CPO/NPO/LPO	Experimental/roadmap stage	Needs complete matched system proof	Depends on electrical boundary and packaging maturity	Advanced pluggables retain comparable delivered economics
Accelerator optical I/O	Strategic prototype/roadmap stage	Needs customer product evidence	Separate scale-up adoption curve	No integration or customer-volume confirmation

7.4 Scenario conditions

Bear case: advanced pluggables, LPO, AECs and NPO close enough of the power and electrical-margin gap that operators preserve modular replacement. CPO remains a selective system feature, with high integration burden and modest external component economics.

Base case: switch-side 200G CPO obtains repeat production evidence during the 2026–2027 verification window. Adoption remains concentrated in the highest-bandwidth fabrics because the operational and supply-chain requirements are demanding. This is the report's central timing view, not a forecast of CPO market share.

Bull case: final-engine yield, external-light/service architecture and qualified manufacturing become repeatable across leading platforms, making CPO a standard option at the high end of Ethernet scale-out and extending the design logic into selected scale-up systems.

Market-research forecasts should not be used to resolve this question. Public reports publish CPO market values with materially different starting sizes and growth rates, indicating incompatible product boundaries and definitions (ResearchAndMarkets 2026; Mordor Intelligence 2025). A bottom-up product-and-content bridge is therefore more useful than averaging top-down estimates.

7.5 What counts as a timing miss

The report will treat a delayed roadmap differently from a disconfirmed architecture. A schedule slip without negative qualification evidence can move the commercial-proof window but leave the technical thesis intact. A repeated inability to identify a customer CPO SKU, accepted-unit range or repeat shipment after a claimed production period weakens the commercial interpretation. A field or qualification issue that requires a return to a pluggable or NPO boundary is stronger disconfirming evidence because it attacks the architecture's operational premise.

Event	Timing interpretation	Investment read-through
Product sampling	Early route validation	Not commercial proof
Vendor says production	Positive timing signal	Requires customer/quantity corroboration
Named customer deployment	Stronger system signal	Still needs CPO-SKU and repetition boundary
Second order / multi-customer use	Commercial-proof evidence	Begins adoption confidence
Yield/service disclosure	Manufacturing/economic evidence	Needed for profit-pool attribution
Margin/content disclosure	Earnings evidence	Needed for company materiality

This sequence is deliberately more conservative than a technology-market forecast. It avoids declaring adoption when the public evidence might only show an engineering sample, a platform family, a partner relationship or a programme award. It also makes the report useful on a quarterly basis: each new disclosure can be placed on a clear progression rather than absorbed into a vague "CPO is happening" narrative.

8 Industry structure and value pool

8.1 CPO redistributes content; it does not automatically create value

Layer	Potential CPO effect	Key question
Switch silicon and SerDes	Greater system integration and package co-design	Does ASIC control capture the system benefit?
Optical DSP / pluggable module	Some content may be reduced or relocated	Is it displaced, or reintroduced elsewhere?
PIC/EIC	More design and integration content	Who owns the qualified engine design?
Laser / ELS	Potentially higher strategic importance	Is laser content external, differentiated and margin-accretive?
Packaging / fibre / connector	More complex assembly and test	Can suppliers earn returns above capital and warranty cost?
Test equipment	More insertion points across wafer/package/system	Does test complexity create durable pricing power?

8.2 Profit-pool test

The supplier most likely to capture economics is not necessarily the one with the best device. Durable capture requires control of a bottleneck, credible switching costs, customer qualification, and economics that survive yield, capex, rework and warranty.

Inference: In the current public record, the likely control points are platform integration, optical-engine/PIC design, lasers and advanced package/test process control. None has yet established a public CPO-specific profit-pool lead.

8.3 Content migration is not profit creation

CPO can reduce or relocate the conventional module/DSP bill of materials while increasing optical-engine, laser, package, fibre-interface and test content. That does not tell an investor who captures incremental gross profit. A component can be technically indispensable but economically weak if it is qualified at multiple suppliers, subjected to price-down or burdened with capital, yield and warranty costs. Conversely, a relatively small process step can have strategic value if it is scarce, qualified and expensive to replace.

Layer	Candidate controllers	Potential upside	Principal economic leak
Platform and switch ASIC	NVIDIA, Broadcom	System architecture and customer route	Optics value flows to external supply chain
PIC/EIC engine	Coherent, TSMC, specialist SiPh routes	Integration IP and qualified design	Packaging/test burden or multi-source pricing
External laser	Lumentum, Coherent and others	High-power, serviceable optical content	Commodity qualification or low content per system
Advanced package / OSAT	TSMC, OSATs and system integrators	Integration and yield control	Capital intensity, rework and warranty
Fibre/connector/test	Specialist suppliers	Process lock-in and serviceability	Cost-down and limited share of total BOM

8.4 The profit-pool scorecard

For every company, the report asks six questions: does it control a product-critical layer; is that layer qualified in a named system; is it difficult to replace; is supplier share identifiable; can it earn a margin after yield/rework/warranty; and is the result material to consolidated earnings? A "yes" only to the first question is technology optionality, not a completed investment case.

NVIDIA's and Broadcom's announcements establish platform/product control points (NVIDIA 2026c; Broadcom 2025e). NVIDIA's partnerships with Coherent and Lumentum strengthen the supply-chain signal, not the supplier-allocation case (NVIDIA and Coherent 2026; Lumentum 2026c).

9 Company investment cases

9.1 How to read these cases

These are evidence-gated thematic views, not price targets. "Positive" means the company has credible strategic exposure to a potential CPO control point; it does **not** mean CPO earnings are presently quantifiable or that the security is suitable for any investor.

9.2 Expectations versus variant view

This public edition separates three things that are often blurred in technology investing:

1. **Observed public facts** — product, customer, qualification, shipment and filing records linked in the report.
2. **External expectations** — management targets and market estimates. Restricted sell-side research is used privately only; exact proprietary estimates and analyst identities are not published here.
3. **Nur Alpys' variant view** — a derived positive, neutral or negative stance, with a stated catalyst and falsification condition.

The private model is being prepared to translate reconciled consensus baselines into bear/base/bull CPO revenue, EPS and valuation sensitivity. No CPO-specific sensitivity is published until the underlying inputs are sourced, date-stamped and internally reconciled. This is deliberately not a target-price report.

Required input	Public status	Publication rule
Baseline consensus revenue, EPS, capex and valuation	Restricted-source intake pending	Publish only derived ranges where permitted
CPO systems, adoption and engines/system	Partially evidenced	Label scenario assumptions clearly
Supplier content, share, ASP and product margin	Not publicly attributable	Do not infer from consolidated results
Yield, rework, warranty and cannibalisation	Not publicly attributable	Keep as private sensitivity inputs, never facts

Company	Current view	CPO/control route	Core evidence	Main missing proof
NVIDIA	Positive strategic exposure	Integrated Spectrum-X Ethernet Photonics platform	CPO-based 200G-SerDes system described as in production (NVIDIA 2026c)	Units, customer mix, CPO economics and supplier split
Broadcom	Positive enabling exposure	Merchant switch ASIC + third-generation CPO route	200G/lane CPO announcement; 102.4T CPO path (Broadcom 2025b, 2026)	Customer deployment, yields and field data
Coherent	Constructive component exposure	Lasers, SiPh, VCSEL, InP and packaging	Multiple CPO demonstrations (Coherent 2026c)	Design wins, share and CPO margins
Lumentum	Constructive/watch	High-power and DWDM external-laser inputs	OFC 2026 CPO-oriented laser demonstrations (Lumentum 2026b)	Customer conversion, volume and margin attribution
Marvell	Strategic watch	Scale-up optical-I/O/Photonic Fabric	Celestial AI acquisition and scale-up route (Marvell Technology 2026; Marvell 2025)	Production/customer validation and economics
TSMC	Strategic infrastructure watch	Silicon photonics and advanced packaging	Annual-report CPO/silicon-photonics development disclosure (TSMC 2026b)	Merchant economics and customer content capture

9.3 NVIDIA

Fact: NVIDIA states that Spectrum-X Ethernet Photonics combines CPO with 200G SerDes and is now in production (NVIDIA 2026c).

Inference: NVIDIA has the strongest disclosed integrated platform route to commercial switch-side CPO. Its power is system control: ASIC, NIC/DPU, software, reference design and customer route. The public record does not allocate CPO revenue, gross margin or component sourcing to NVIDIA.

Invalidation trigger: disclosed deployments show the CPO element is non-material, yields/service cost are poor, or alternative optics retain comparable system economics.

9.3.1 Product boundary and public evidence

NVIDIA's relevant CPO exposure is not a generic optics claim. It is an integrated networking route: switch silicon and 200G SerDes, optical-engine architecture, software and systems integration, plus a customer/ecosystem route. NVIDIA's materials describe Spectrum-X Ethernet Photonics as a CPO system, while its Vera Rubin production communication places photonics within a broader AI-factory platform (NVIDIA 2026c, 2026b). This is stronger evidence than a component demonstration because it joins the architecture to an announced platform. It is still not a disclosed CPO bill of materials, accepted-unit range or CPO revenue line.

9.3.2 Variant view

Stance: positive strategic exposure; medium confidence. NVIDIA is best positioned to force a CPO architecture decision because it can coordinate the ASIC, NIC/DPU, rack design, software and customer deployment. That coordination can matter more than ownership of any individual optical component. The differentiated question is whether the system benefit is enough to accelerate high-end CPO adoption before alternate pluggable or LPO routes preserve the same economics.

9.3.3 Earnings relevance and falsification

CPO is unlikely to be separately material to NVIDIA's consolidated earnings in the near term unless it expands platform differentiation, system ASP, attach or customer lock-in. The report therefore does not model direct CPO EPS. A positive update requires named CPO SKU/customer evidence and repeat deployment; a negative update follows if the announced platform is deployed primarily with pluggable configurations, or if service/yield evidence shows that the integrated route cannot scale economically. NVIDIA's strategic partnership with Coherent confirms supply-chain importance, not capture of supplier or NVIDIA gross margin (NVIDIA and Coherent 2026).

9.4 Broadcom

Fact: Broadcom announced its third-generation 200G/lane CPO technology and, at OFC 2026, presented a 102.4T CPO route (Broadcom 2025b, 2026).

Inference: Broadcom has the strongest merchant CPO product-definition position. That does not prove that it owns the engine profit pool, that customers deploy at volume, or that merchant CPO earns structurally higher returns.

9.4.1 Product boundary and public evidence

Broadcom provides the cleanest merchant-switch product definition in the report. Its Tomahawk 6 Davisson materials identify a 102.4T Ethernet switch/CPO route and 200G SerDes; the product catalogue separately shows CPO as a defined product family (Broadcom 2025e, 2025d). Broadcom also announced a third-generation CPO technology with 200G/lane capability (Broadcom 2025b). These are credible indicators that a merchant platform vendor is spending product resources on the transition.

9.4.2 Variant view

Stance: positive enabling exposure; medium confidence. Broadcom's potential advantage is architectural leverage: a merchant switch ASIC and a specified CPO platform can influence engine interfaces, ecosystem qualification and customer reference designs. That makes it a likely enabler even if it does not manufacture every engine, laser or package. The risk is that the optical engine becomes externally sourced pass-through content while system integration, service and yield complexity slow customer adoption.

9.4.3 Earnings relevance and falsification

CPO's incremental effect on Broadcom depends on whether it protects switch-silicon share, increases platform value, lowers customer power constraints or creates new high-margin content. Its contribution cannot be inferred from overall semiconductor revenue or corporate margin. The stance is strengthened by named customer deployments, repeated shipments and evidence that Broadcom controls differentiated CPO silicon/content. It is weakened if customers use the platform mainly as a technology evaluation route, if pluggable/LPO alternatives meet the same 200G system requirements, or if external suppliers capture the incremental economics.

9.5 Coherent and Lumentum

Fact: Coherent demonstrated socketed CPO, external-laser, VCSEL and InP-on-silicon routes at OFC 2026; Lumentum demonstrated high-power CPO-oriented laser technology and a 16-channel DWDM laser source (Coherent 2026c; Lumentum 2026b).

Inference: Both are credible optical-content beneficiaries if external laser and optical-engine content expands. Coherent's disclosed technology breadth is broader; Lumentum's laser route is more focused. Neither has public CPO-specific revenue, qualified-engine yield or gross-margin attribution.

9.5.1 Coherent: broad optical-engine option value

Stance: constructive component exposure; medium confidence. Coherent's public CPO evidence spans silicon photonics, external lasers, VCSELs, InP modulators and packaging-oriented routes (Coherent 2026c, 2026d). That breadth is valuable because the winning scale-out engine may use a different PIC/laser/service configuration than early demonstrations. Its central risk is equally clear: a broad technology stack does not identify the customer product, supplier share, qualified manufacturing yield or CPO-specific margin.

The variant thesis is that breadth plus NVIDIA partnership/capacity alignment can create a durable role in a high-value engine or laser layer. Evidence that would make this investable is a named CPO design win, defined content boundary, conversion into repeat production and a margin discussion that distinguishes AI optics from consolidated company performance. The case is invalidated if the company remains a multi-source technology participant, if qualified CPO content is lower than expected, or if the cost of external-light and package integration consumes the theoretical value.

9.5.2 Lumentum: external-laser conversion case

Stance: constructive/watch; medium confidence. Lumentum's CPO relevance is more concentrated in high-power laser, external-light and DWDM laser routes. Its OFC material identifies technologies aimed at scale-out and scale-up optical infrastructure (Lumentum 2026b). The announced NVIDIA optics partnership and new U.S. laser manufacturing facility strengthen the strategic/capacity signal (Lumentum 2026c, 2026a). They do not disclose CPO units, customer-specific laser content, realised price or gross margin.

The differentiated question is whether external light becomes a serviceable, scarce and margin-accretive CPO layer rather than a qualified component with modest content and price-down exposure. Positive evidence would be conversion of a disclosed order or capacity commitment into named product shipments and segment-level economics. Negative evidence would be multi-source commodity pricing, lower-than-expected laser count/content, or a system architecture that integrates light differently.

9.6 Marvell and TSMC

Fact: Marvell completed the Celestial AI acquisition and describes Photonic Fabric as a scale-up optical-interconnect route; TSMC's annual report says it is developing silicon photonics and CPO solutions (Marvell Technology 2026; TSMC 2026b).

Inference: Marvell is the strongest public-company vehicle for accelerator optical-I/O optionality, a distinct domain from switch-side CPO. TSMC is a likely manufacturing/packaging control point, not a directly comparable optical-engine supplier.

9.6.1 Marvell: a separate scale-up optionality case

Stance: strategic watch; low-to-medium confidence. Marvell's Celestial AI acquisition and its silicon-photonics light-engine work place it at the accelerator optical-I/O and rack-scale interconnect boundary rather than squarely in the switch-CPO comparison (Marvell Technology 2026; Marvell 2025). This can be strategically important because scale-up fabrics may require tighter electrical/optical integration than conventional Ethernet scale-out. But it has a different customer denominator, software/system dependency, product cycle and service model.

The report will not use switch-CPO production signals as evidence for Marvell's accelerator optical-I/O revenue. The stance improves with a named product, customer adoption, realised Photonic Fabric revenue or a disclosed content/margin bridge. It weakens if the integration remains a roadmap/demo, if competing interconnect standards capture the relevant deployments, or if the acquired technology adds capability without commercial scale.

9.6.2 TSMC: process control, not an optical-engine peer

Stance: strategic infrastructure watch; medium confidence. TSMC's CPO relevance lies in silicon photonics, heterogeneous EIC/PIC integration and advanced packaging, not in acting as a merchant optical-engine supplier. Its technology symposium and annual report establish that these capabilities are active development areas (TSMC 2024, 2026b). The potential strategic value is process control: if CPO packages require a scarce integration flow, the foundry/package route can shape what platforms become manufacturable.

The investment limitation is attribution. Advanced packaging and silicon photonics can be valuable without being separately visible in TSMC revenue, margin or customer disclosures. A stronger case would require programme-level production volume, a clearer technology/service boundary or signs of constrained capacity and pricing power. It is weakened if CPO remains low volume, if integration is widely reproducible, or if the value concentrates in system suppliers rather than the manufacturing stack.

9.7 What the consensus-versus-variant framework can and cannot do today

Consolidated estimates are only a scale check: diversified platform companies may capture CPO through system differentiation, while optical suppliers still lack attributable content, margin and volume data. The report requires both a proven product/control point and a reconciled revenue-and-margin bridge; no company clears the second test today.

Input needed for a CPO earnings bridge	NVIDIA	Broadcom	Coherent	Lumentum	Marvell	TSMC
Named product and route	Partial	Strong	Partial	Partial	Partial	Partial
Named customer CPO SKU / volume	Open	Open	Open	Open	Open	Open
Supplier content / qualified share	Open	Open	Open	Open	Open	Open
CPO-specific ASP / gross margin	Open	Open	Open	Open	Open	Open
Yield / rework / warranty allocation	Open	Open	Open	Open	Open	Open
Defensible public EPS sensitivity	No	No	No	No	No	No

9.7.1 How a future model will be constructed

For a platform or component supplier, incremental CPO revenue must be built from a common unit chain:

$$\text{CPO revenue} = \text{systems} \times \text{adoption} \times \text{engines/system} \times \text{content/engine} \times \text{qualified share.}$$

Incremental gross profit must then subtract the correct product cost, yield/rework burden, warranty and cannibalised legacy content. The model must keep supplier layers mutually exclusive. For example, an optical engine cannot be counted once as Broadcom platform value, again as Coherent PIC value, again as Lumentum laser value and again as TSMC packaging value unless the separate transfer-price/content boundaries are identified. That is why the report publishes relative stances rather than a spurious market-size allocation.

9.7.2 Catalysts by company

NVIDIA: a customer-confirmed Spectrum-X Photonics CPO SKU with a repeat order and a clear distinction from pluggable SN6600-LD configurations. The product-family boundary matters more than an undifferentiated platform-adoption headline.

Broadcom: a named Tomahawk 6 CPO deployment with an integrator/customer, production sequence and a service/qualification statement. The existing product definition is strong; the missing catalyst is external proof of durable adoption.

Coherent: a programme-level optical-engine or external-light design win and evidence that its broad device portfolio converts into qualified revenue rather than optionality alone.

Lumentum: conversion of strategic-capacity/laser positioning into disclosed CPO-adjacent demand, content and margin logic. A capacity announcement is a supply signal, not proof of economically attractive utilisation.

Marvell: a customer/product announcement that makes the accelerator optical-I/O route commercial rather than architectural. The important validation is scale-up fabric economics, not switch-CPO extrapolation.

TSMC: a public programme that links its photonics/advanced-packaging process to production throughput or constrained capacity. A technology demonstration demonstrates capability, not captured value.

Why there are no target prices

The public record does not support a defensible CPO-specific revenue, margin, diluted-share or valuation bridge for any of the six companies. Publishing a target price would hide the most important uncertainty: which supplier owns a qualified, profitable portion of the engine and system stack. The correct output is a relative stance with evidence gates.

10 Risks, disconfirming evidence and watchlist

10.1 Thesis risks

Risk	What it would mean	Evidence to monitor
Pluggable/LPO innovation	CPO is not required for major scale-out links	Matched 200G/400G lane-rate power, BER and field data
Yield and qualification	CPO cannot turn prototype integration into qualified output	Accepted yield, rework, Cpk, qualification and field-return data
Serviceability	Operators reject deeply integrated failure domains	Replacement procedures, MTTR and warranty allocation
Weak value capture	CPO shifts technical work without creating supplier margins	ASP, supplier share, gross margin and price-down evidence
Customer concentration	One platform's success does not generalise to an industry	Named SKU, customer count, repeat-shipment and port data

10.2 Quarterly watchlist

1. Production CPO SKU, named customer and accepted-unit/port range.
2. Repeat shipment or second-customer evidence.
3. Final-engine yield waterfall: fibre attach, package, test/rework and customer acceptance.
4. Qualification and field-service record.
5. Matched CPO/LPO/NPO total-cost comparison.
6. Supplier content, ASP, gross margin, capex and warranty allocation.

10.3 Re-rating rule

The report will not upgrade a technology demo to a profit-pool conclusion. A material upgrade requires evidence from both a **commercial/system** gate (customer, units, deployment) and an **economic** gate (content, yield, margin or warranty) in the same product boundary.

10.4 Evidence that would change the ranking

Observation	Read-through	Action
Customer confirms CPO SKU, accepted unit range and repeat order	Clears a commercial proof gap	Upgrade timing confidence; do not infer supplier economics
Supplier identifies qualified content, share or transfer-price boundary	Begins profit-pool attribution	Assess materiality against company revenue and margin
Final-engine yield/rework or field-return metric becomes public	Tests whether integration is economically viable	Revisit CPO versus serviceable alternatives
LPO or pluggable route demonstrates matched power, reach and service cost	Direct architecture counterevidence	Downgrade CPO necessity at that boundary
CPO announcement is tied only to lab, evaluation or a pluggable SKU	Weakens inferred deployment	Withdraw or narrow commercial interpretation

10.5 Quarterly evidence dashboard

The next edition should answer the same questions with a dated before/after comparison. The watchlist is intentionally short enough to use:

1. Which exact CPO SKU is shipping, to whom, in what range, and is the shipment repeated?
2. What is the optical-engine and external-laser boundary in that SKU?
3. Is a yield, qualification or service metric disclosed after final package integration?
4. Has an LPO/NPO/pluggable counterexample closed the relevant system gap?
5. Has company guidance, backlog, capex or margin commentary made the CPO contribution more or less material?

> ⚠ Current limiting factor
>
> The report has enough evidence to rank technical routes and commercial signals. It does not have enough public evidence to calculate CPO-specific supplier revenue, gross margin, EPS or target prices. That is an evidence boundary, not a missing spreadsheet cell.

The main analytical risk is narrative leakage: treating a platform announcement as a CPO-SKU deployment, a partnership as supplier allocation, a component result as production yield, or a market forecast as company revenue. The next update withdraws an inference whenever a newly disclosed product boundary requires a narrower reading.

A Methodology and public-data boundary

A.1 Method

The report works from the following chain:

$$\text{technical constraint} \rightarrow \text{architecture choice} \rightarrow \text{supplier content} \rightarrow \text{qualified economics} \rightarrow \text{investment outcome}$$

Each link must be evidenced separately. A conclusion is downgraded when its support depends on an unobserved link.

A.2 What is public

- Original, publicly linkable sources and attributable public statements.
- Original analytical diagrams, tables and scenarios.
- A public bibliography and source/figure register.

A.3 What remains private

- Copyrighted PDFs and raw archives.
- Private reading notes, interview notes, contacts and unpublished assumptions.
- Working models and source files that cannot be reproduced from public evidence.
- Unverified or incomplete claims.
- Licensed analyst reports, exact consensus estimates, detailed model pages and analyst identifiers unless their licence terms explicitly permit publication.

A.4 Release checklist

Before a quarterly edition is published: reconcile claims with the internal evidence record; verify citations and source links; review every figure for attribution and public-use basis; label facts/inferences/forecasts; record changed and withdrawn conclusions; and render the HTML and PDF outputs.

A.5 Research protocol

The project follows a falsification-first discipline. A conclusion is not upgraded because an additional source repeats an attractive narrative. It is upgraded when new evidence clears a decision gate that was previously open. For switch-side CPO, the relevant gates are: a defined architecture and SKU; qualified package/manufacturing evidence; customer acceptance and repeated shipment; and attributable supplier economics. The same logic applies to scale-up optical I/O, but the system, customer and component boundaries are different.

This matters because CPO has a natural evidence asymmetry. Product companies can legitimately disclose architecture, roadmaps, port counts, demonstrations and ecosystem partners long before they can disclose customer volumes, yield, cost, service history or supplier allocations. Academic research can demonstrate an electrical, optical or package sub-boundary with great precision, yet still have no evidence about final-engine cost. The methodology preserves that asymmetry rather than treating all positive evidence as equivalent.

A.5.1 Source classification

Source class	What it can establish	What it cannot establish without corroboration
Customer/operator disclosure	Product use, named deployment, operating context	Supplier share, accepted-unit count or margin unless stated
Audited filing	Consolidated financial facts, risks, capital intensity	CPO revenue/margin allocation unless separately disclosed
Company product material	Roadmap, stated architecture, product specification	Independent field performance, market-wide adoption
Standards / implementation agreement	Interface requirements and defined boundaries	Product economics or customer volume
Peer-reviewed paper	Measurement at the reported test boundary	Production yield, warranty, field deployment or profit pool
Consultant / market report	External forecast framing	A reconciled product denominator or company attribution

A.5.2 Reconciliation rules

The report applies five controls before making a comparison. First, compare only the same architecture: CPO, NPO, LPO, retimed pluggable and accelerator optical I/O are not interchangeable labels. Second, identify the physical boundary: lane rate, port count, reach, FEC, electrical loss, laser inclusion and cooling must be stated. Third, identify the commercial boundary: vendor roadmap, customer evaluation, paid shipment and repeat production mean different things. Fourth, identify the financial boundary: consolidated company metrics cannot become CPO product margins. Fifth, identify the source date: current technical claims can be stale in a fast-moving product cycle.

These rules explain why the report sometimes withholds an answer. For example, a production statement can support a timing update but not a supplier earnings bridge if customer SKU, unit, component content and pricing remain undisclosed. A market forecast can describe external expectations but cannot allocate value across a switch platform, PIC, laser, package and test supplier. Withholding an unsupported number is a research conclusion, not an omission.

A.5.3 Scenario methodology

Where public evidence does not support a single forecast, the report uses bear/base/bull conditions. The bear case preserves modular alternatives or exposes yield/service penalties. The base case assumes commercial proof at a defined product boundary but not broad market adoption. The bull case requires repeatable qualified manufacturing, serviceability and multiple-platform expansion. Scenario tables are directional unless their inputs have a dated public source and a stated calculation method.

The report does not convert unknown yield, customer volume or product margin into a "base case" simply because an external analyst or management team is optimistic. External expectations are treated as inputs to a future private sensitivity model. Public output will remain a derived range or qualitative stance unless the number is reproducible from permissible public inputs.

A.5.4 Independence and conflicts

Vendor material establishes the vendor's claim, not a market fact; the report tests it against customer/operator material, standards, filings and independent technical work where available. The author has no disclosed commercial relationship with the companies discussed.

B Glossary

Term	Meaning in this report
CPO	Co-packaged optics: optical engines integrated close to a host ASIC/package boundary
LPO	Linear pluggable optics: pluggable optics with reduced/omitted retiming/DSP functions
NPO	Near-packaged optics: optical engines placed near, but not fully co-packaged with, the host ASIC
PIC	Photonic integrated circuit; performs optical functions such as modulation, multiplexing and detection
EIC	Electronic integrated circuit; includes drivers, TIAs and related electrical functions
ELS	External laser source, physically separated from the optical engine
SerDes	Serializer/deserializer electrical I/O circuitry
TCO	Total cost of ownership, including service, yield and operational cost where measurable
Qualified yield	Accepted output after assembly, test, environmental screening and relevant customer criteria
Retimed pluggable	A modular optical transceiver that contains retiming/DSP functions and is replaced at the faceplate boundary
ELSFP	External Laser Small Form Factor Pluggable; an OIF-defined external-light interface concept
Optical engine	Coupled PIC, EIC, laser-interface, fibre-interface, package and test implementation rather than merely a photonic die
Known-good die	A die tested before high-value final integration; it reduces but does not eliminate final-package yield risk
First-pass yield	Share of units that pass the stated process/test boundary without rework
Rework	Repair, replacement or retest activity performed after an initial assembly/test failure
Accepted unit	A system or engine that passes the defined supplier/customer acceptance boundary; it is stronger evidence than a lab sample

Term	Meaning in this report
Commercial proof	In this report, repeat paid production from two independent customers or sustained/repeated production by one major customer, with the CPO boundary clear
Meaningful adoption	A disclosed unit/port denominator sufficient to assess economic relevance; it is distinct from a single production claim
Content	The dollar or functional portion of a product supplied by a company; it is not assumed from a partner list
Qualified share	The fraction of a defined product boundary supplied by a company after qualification; it must not be inferred from ecosystem membership
CPO product boundary	The exact switch/accelerator SKU, optical configuration, architecture and relevant service boundary to which a claim applies
Scale-out	Network expansion across many nodes/racks, typically prioritising flexibility, topology and operational serviceability
Scale-up	High-bandwidth interconnect inside or among tightly coupled compute domains; it can have a different optical-I/O architecture and buyer
Facility-adjusted power	Optical-interface power plus explicitly stated conversion/cooling assumptions; it is not complete data-centre energy without full-system inputs
Falsification trigger	Evidence that would materially weaken or invalidate a stated inference or forecast

C Sources and figure register

C.1 Source policy

The bibliography contains the public source of record. This website does not host source PDFs or raw web-page archives. Links open the original publisher, investor-relations or standards location.

C.2 Figure register

Figure	Status	Source and calculation boundary
1. Signal-path comparison	Original	Conceptual CPO versus pluggable path; not to scale
2. Architecture decision tree	Original	Analytical decision framework; not a market-share forecast
3. Optical-engine control stack	Original	PIC/EIC/laser/package/test framework; no supplier allocation implied
4. Qualified-good-engine waterfall	Original	Manufacturing framework; no supplier yield implied
5. Adoption gates	Original	Commercial-proof methodology; no market-share forecast
6. Value-pool migration	Original	Analytical framework; revenue and margin attribution remain open
7. CPO earnings bridge	Original	Formula/control framework; no EPS sensitivity is calculated
8. Physical-problem diagram	Original	Conceptual CPO signal-path comparison
9. Engine-system diagram	Original	Conceptual engine co-design diagram
10. Profit-pool chain	Original	Analytical value-capture framework
Retained-paper snapshots	Not published	Source figures remain outside the public site pending copyright and editorial review

C.3 Reproducibility note

Every quantitative table identifies whether it is a public fact, a calculation, an assumption-labelled scenario or an open evidence gap. Original diagrams communicate the report's structure; they do not recreate copyrighted source figures. The decision evidence register identifies the public source and limitation for each selected evidence record.

C.4 Link integrity and archival policy

Public links point to the original publisher, investor-relations, standards or DOI location. A link can fail or change after publication; the next quarterly update will either replace it with a stable official location, withdraw the dependent assertion, or retain the record as historical with an explicit availability note. The public site intentionally does not mirror third-party PDFs or pages.

C.5 Bibliography

D Decision evidence register

D.1 Purpose and reading rule

This register is the public release manifest for the first edition. It selects 80 decision-relevant evidence records from the wider research system. Each record is deliberately narrow: it tells the reader what the linked public record establishes, where it is used, and what it does **not** establish. The register is not a list of 80 conclusions. It is an audit trail for the conclusions elsewhere in the report.

The source key is an original public publication, filing, standards document, official product material, customer/operator statement or peer-reviewed paper. “Fact” means the source’s direct statement or reported measurement. “Inference” means the report’s interpretation. “Forecast” means a conditional timing view. No restricted report, private source archive, unpublished model or source-PDF snapshot is included.

D.2 Architecture and electrical boundary

ID	Public evidence record	Type	Used in	Boundary / limitation
P-01	OIF identifies power and signal-integrity pressure at energy-sensitive interfaces (Optical Internetworking Forum 2024).	Fact	1, 2	Requirements document; not a product comparison
P-02	OIF documents a 3.2T co-packaged-module implementation framework (Optical Internetworking Forum 2023).	Fact	2, 3	Industry framework; not volume evidence

ID	Public evidence record	Type	Used in	Boundary / limitation
P-03	IEEE 802.3dj provides the public lane-rate standards process (IEEE 802.3 2024).	Fact	1, 3	Standards process; not adoption proof
P-04	Broadcom describes a 200G/lane third-generation CPO route (Broadcom 2025b).	Fact	3, 8	Vendor product claim
P-05	Broadcom describes a 102.4T Davisson CPO switch route (Broadcom 2025e).	Fact	3, 8	Product definition, not accepted-unit disclosure
P-06	Broadcom lists CPO switches as a product category (Broadcom 2025d).	Fact	3, 8	Catalogue presence, not customer deployment
P-07	NVIDIA describes CPO power-efficiency logic for AI factories (NVIDIA 2025b).	Fact	2	Vendor technical framing
P-08	NVIDIA describes cross-industry CPO collaboration and engine architecture (NVIDIA 2025a).	Fact	2, 4	Vendor architecture statement
P-09	Zhou et al. model 400G/lane linear-drive constraints (Zhou et al. 2026).	Fact	3	Model boundary, not a shipped system

ID	Public evidence record	Type	Used in	Boundary / limitation
P-10	The report infers that 200G/lane is the strongest switch-CPO public signal.	Inference	Executive summary	Requires future customer/volume validation

D.3 System power and alternative architectures

ID	Public evidence record	Type	Used in	Boundary / limitation
P-11	Marvell demonstrates a silicon-photonics light engine for rack-scale interconnect (Marvell 2025).	Fact	2, 3, 8	Vendor demo; not switch-CPO share
P-12	Oracle presents Acceleron as a cloud-networking route (Oracle 2026).	Fact	2, 6	Operator/product countercase; not matched CPO TCO
P-13	OIF's energy work supports a delivered-bit comparison framework (Optical Internetworking Forum 2024).	Inference	2	Does not set a universal pJ/bit winner
P-14	The 102.4T power table is an explicit scenario calculation.	Forecast	2	Assumption-labelled; not measured chassis power
P-15	The central scenario favours CPO strongly versus fully retimed interfaces.	Forecast	2	Depends on stated per-port assumptions

ID	Public evidence record	Type	Used in	Boundary / limitation
P-16	The same scenario shows a much narrower CPO advantage versus LPO.	Forecast	2	Does not include field service or matched chassis evidence
P-17	Retimed pluggables retain modular replacement advantages.	Fact	1, 3	Architecture principle; economics vary by system
P-18	LPO can be a serious countercase where host margin is adequate.	Inference	2, 3	Requires matched product evidence
P-19	NPO may provide an intermediate electrical/service boundary.	Inference	3	Product-specific route remains open
P-20	Accelerator optical I/O is a distinct adoption/value-pool domain.	Inference	1, 3, 8	Must not be aggregated with switch CPO

D.4 Optical engines, PICs and external light

ID	Public evidence record	Type	Used in	Boundary / limitation
P-21	Coherent shows multiple CPO technologies at OFC 2026 (Coherent 2026c).	Fact	2, 4, 8	Technology breadth, not design-win proof
P-22	Coherent's investor event describes its technology portfolio (Coherent 2026d).	Fact	2, 4, 8	Company presentation; no supplier allocation

ID	Public evidence record	Type	Used in	Boundary / limitation
P-23	TSMC's symposium documents silicon-leadership and AI technology direction (TSMC 2024).	Fact	4, 8	Capability signal, not CPO revenue
P-24	TSMC's annual report documents the company context (TSMC 2026b).	Fact	4, 8	Consolidated disclosure; no CPO margin line
P-25	OIF ELSFP documents an external-laser interface framework (Optical Internetworking Forum 2025).	Fact	4, 5	Standard/interface; not service-cost proof
P-26	Lumentum shows CPO-oriented optical infrastructure technologies (Lumentum 2026b).	Fact	4, 8	Vendor demonstration
P-27	Lumentum/NVIDIA announce an optics technology partnership (Lumentum 2026c).	Fact	7, 8	Partnership, not content allocation
P-28	NVIDIA/Coherent announce an optics technology partnership (NVIDIA and Coherent 2026).	Fact	7, 8	Partnership, not realised supplier margin

ID	Public evidence record	Type	Used in	Boundary / limitation
P-29	Coherent's multi-route portfolio creates engine optionality.	Inference	4, 8	Requires design-win and yield proof
P-30	External light can improve service separation but creates its own qualification work.	Inference	4, 5	Not a universal serviceability solution

D.5 Manufacturing, qualification and serviceability

ID	Public evidence record	Type	Used in	Boundary / limitation
P-31	Silicon-photonics test creates new production-test challenges (Teradyne 2025).	Fact	5	Supplier perspective; no platform test-time data
P-32	CPO qualified cost must include assembly, test, rework, capital and warranty.	Inference	5	Formula; not a supplier cost disclosure
P-33	Multi-interface yield compounds across an engine or board.	Fact	5	Mathematical relationship; interface independence is illustrative
P-34	1,024 interfaces at 99.9% assumed connection yield produce 35.9% all-good yield.	Forecast	5	Illustrative calculation, not supplier yield
P-35	95% board yield at 1,024 interfaces requires ~99.995% per-connection yield.	Forecast	5	Illustrative calculation, no rework model

ID	Public evidence record	Type	Used in	Boundary / limitation
P-36	A detachable laser boundary does not automatically replace an optical engine.	Inference	5	Needs product-specific service procedure
P-37	Field replacement must be assessed through MTTR, spares and warranty.	Inference	5	Public field data remain limited
P-38	Final-engine yield is a decision-critical missing data set.	Open question	5, 9	No broad public disclosure identified
P-39	Qualification after final integration matters more than a component-only result.	Inference	5	Product boundary required
P-40	No public source in this edition establishes a comparable CPO field-return distribution.	Open question	5, 9	Evidence gap

D.6 Customer, product and commercial-proof signals

ID	Public evidence record	Type	Used in	Boundary / limitation
P-41	NVIDIA states Spectrum-X Ethernet Photonics is in production (NVIDIA 2026c).	Fact	Executive, 3, 6, 8	Vendor production statement; units not disclosed

ID	Public evidence record	Type	Used in	Boundary / limitation
P-42	NVIDIA describes Vera Rubin platform production context (NVIDIA 2026b).	Fact	Executive, 8	Broad platform context
P-43	Broadcom presents a 102.4T CPO route at OFC 2026 (Broadcom 2026).	Fact	Executive, 3, 8	Event material; no customer volume
P-44	Celestica reports a CPO Ethernet-switch programme and expected ramp (Celestica 2026b).	Fact	6, 7	Customer/content and realised volume remain open
P-45	Celestica presents DS6000-series 1.6TbE switches (Celestica 2026a).	Fact	6	Configuration/CPO boundary requires care
P-46	NVIDIA production language is stronger than a lab demo but weaker than customer-confirmed volume.	Inference	Executive, 6	Await repeat shipment record
P-47	A named SKU plus accepted units and repetition defines commercial proof in this report.	Forecast	6, 9	Research convention
P-48	Production claims without a unit denominator cannot establish market leadership.	Inference	Executive, 6	Customer evidence required

ID	Public evidence record	Type	Used in	Boundary / limitation
P-49	The first deployment domain is inferred to be Ethernet scale-out switching.	Inference	Executive, 6	Scale-up remains a separate path
P-50	2026–2027 is the commercial-proof verification window.	Forecast	Executive, 6	Conditional timing view

D.7 Adoption scenarios and counterevidence

ID	Public evidence record	Type	Used in	Boundary / limitation
P-51	Advanced pluggables remain a viable architecture baseline.	Inference	3, 6	Must be compared at same system boundary
P-52	LPO must show qualified system evidence, not component power alone.	Inference	3, 6	Evidence gate
P-53	400G/lane raises electrical-boundary pressure.	Inference	3, 6	Does not establish CPO economics
P-54	400G/lane system proof remains incomplete in public evidence.	Open question	3, 6	Component/model evidence exists
P-55	Bear case: modular alternatives retain adequate system economics.	Forecast	6	Scenario condition

ID	Public evidence record	Type	Used in	Boundary / limitation
P-56	Base case: 200G switch CPO gains repeat production proof.	Forecast	6	Requires named customer and repeat shipments
P-57	Bull case: yield and serviceability become repeatable across platforms.	Forecast	6	Requires production economics
P-58	Top-down CPO market forecasts are too definition-sensitive to average.	Inference	6	Commercial reports use different boundaries
P-59	ResearchAndMarkets publishes a CPO market forecast (ResearchAndMarkets 2026).	Fact	6	Paid-report excerpt; methodology not public
P-60	Mordor publishes a CPO market forecast (Mordor Intelligence 2025).	Fact	6	Paid-report excerpt; definition differs

D.8 Value-pool and company exposure

ID	Public evidence record	Type	Used in	Boundary / limitation
P-61	Platform control can shape architecture without owning all optical profit.	Inference	7, 8	Supplier allocation missing
P-62	PIC/EIC integration can be a potential control point.	Inference	4, 7	Needs qualified share and cost evidence
P-63	External lasers can be strategic CPO content.	Inference	4, 7, 8	Requires content and margin attribution

ID	Public evidence record	Type	Used in	Boundary / limitation
P-64	Package/test can be a control point but can also absorb yield/warranty costs.	Inference	5, 7	Economics not public
P-65	NVIDIA has the clearest integrated platform route.	Inference	Executive, 8	Incremental economics unallocated
P-66	Broadcom has the clearest merchant CPO product definition.	Inference	Executive, 8	Customer deployment and margin open
P-67	Coherent has broad disclosed component-stack exposure.	Inference	Executive, 8	Design win/share/margin open
P-68	Lumentum has a focused external-laser conversion case.	Inference	Executive, 8	CPO units and economics open
P-69	Marvell has accelerator optical-I/O optionality distinct from switch CPO.	Inference	3, 8	Commercial product proof open
P-70	TSMC is a process/integration control-point case.	Inference	4, 8	Attributable economics open

D.9 Investment conclusion and falsification

ID	Public evidence record	Type	Used in	Boundary / limitation
P-71	No public equity has a proven CPO-specific profit-pool lead.	Open question	Executive, 7, 8	Supplier share/yield/margin not disclosed

ID	Public evidence record	Type	Used in	Boundary / limitation
P-72	No company has a defensible public CPO EPS sensitivity today.	Open question	8	Inputs are not reconciled
P-73	No target price is published for a CPO thesis.	Fact	8	Deliberate methodology rule
P-74	NVIDIA is positive strategic exposure at medium confidence.	Inference	Executive, 8	Must clear customer/economic gates
P-75	Broadcom is positive enabling exposure at medium confidence.	Inference	Executive, 8	Must clear deployment/economic gates
P-76	Coherent is constructive component exposure at medium confidence.	Inference	Executive, 8	Must clear content/margin gates
P-77	Lumentum is constructive/watch at medium confidence.	Inference	Executive, 8	Must clear conversion/margin gates
P-78	Marvell and TSMC remain strategic watch cases.	Inference	Executive, 8	Product/value attribution incomplete
P-79	A material upgrade requires one commercial and one economic proof point.	Forecast	9	Research re-rating rule
P-80	The next decisive record joins SKU, units, repetition, content and yield/service evidence.	Open question	Executive, 9	Highest-value evidence gap

D.10 Publication controls

This register is reviewed at each quarterly edition. A record can be withdrawn when its source becomes unavailable, superseded when a more direct primary disclosure appears, or narrowed if a product-family/SKU boundary is resolved differently. The report does not preserve an assertion merely because it was favourable to the thesis.

E Change log

E.1 2026 Q3 — v1.0

- Initial public edition.
- Establishes a positive strategic view of NVIDIA and Broadcom's switch-CPO exposure, and constructive optical-content views of Coherent and Lumentum.
- Separates commercial proof from meaningful adoption and declines to assign unsupported CPO market-share or profit-pool percentages.
- Adds a public-data boundary: raw source archives, private research and unverified models are excluded.

E.2 2026 Q3 — v1.1 (framework update)

- Adds an expectations-versus-variant methodology for the six core companies.
- Keeps licensed sell-side reports and exact proprietary estimates private; the public report will publish only permissible derived ranges and conclusions.
- Does not add a numeric CPO EPS, valuation or target-price conclusion before restricted inputs are received and reconciled.

E.3 2026 Q3 — v1.2 (public investment report expansion)

- Expands the public book into an evidence-gated investment report covering architecture, power, optical engines, manufacturing, adoption, value capture, six company cases and a quarterly watchlist.
- Adds a public decision-evidence register with 80 selected records and a curated bibliography of primary public company, customer, standards and technical sources.
- Adds original analytical diagrams and explicit comparison boundaries; source-PDF snapshots and private archives remain excluded.
- Restates the central conclusion: 200G/lane switch CPO has the strongest disclosed commercial signal, while volume, supplier profit-pool leadership and CPO-specific equity earnings remain unproven (NVIDIA 2026c; Broadcom 2025b, 2026).

E.4 Future quarterly updates

Each update will list new sources, changed assessments, claims withdrawn or downgraded, and unresolved evidence gaps. Historical editions will remain identifiable by version and date.

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